PharmAchieve

VIRAL HEPATITIS

Qualifying Exam Preparatory Course

COMPETENCIES COVERED IN THIS PRESENTATION…

Providing Care: Clinical Care	Providing Care: Distribution	Knowledge & Expertise
Communication & Collaboration	Leadership & Stewardship	Professionalism

LEGEND

ALT	Alanine Aminotransferase
AU	Antigen Units
ELISA	Enzyme-Linked Immunosorbent Assay
HAV	Hepatitis A Virus
HBV	Hepatitis B Virus
HCV	Hepatitis C Virus
HDV	Hepatitis D Virus
HEV	Hepatitis E Virus
HIV	Human Immunodeficiency Virus
HBcAg	Hepatitis B core Antigen
HBeAg	Hepatitis B e Antigen
HBsAg	Hepatitis B surface Antigen
Anti-HBsAg	Antibody to HBsAg
SVR	Sustained Virologic Response

OBJECTIVES

1. Explain the pathophysiology of viral hepatitis
2. Understand diagnostic methods and their interpretation
3. Identify the clinical presentation and define acute and chronic hepatitis
4. Identify goals of therapy
5. - Be knowledgeable regarding the management of viral hepatitis which includes:
     - Non-pharmacological alternatives
     - Pharmacological alternatives
6. Monitoring parameters

PATHOPHYSIOLOGY

FYI

Characteristics	HAV	HBV	HCV	HDV	HEV
Virus type	RNA	DNA	RNA	RNA	RNA
Incubation (days)	15-45	30-180	15-60	21-140	15-65
Transmission	Fecal-oral (most common)	Percutaneous Sexual Vertical*	Percutaneous Sexual Vertical*	Percutaneous Sexual Vertical*	Fecal-oral Zoonotic Percutaneous Vertical*
Prevention	Pre/postexposure immunization	Pre/postexposure immunization, maternal screening	Blood donor screening, risk behaviour modification	HBV immunization prevents HDV infection	Ensure safe drinking water, adequately cooked pork

HDV requires coexisting HBV infection
Superinfection: a chronic HBV carrier may be infected with HDV
Co-infection: an individual can be infected by both HDV and HBV at the same time

*Vertical transmission: from mother to child

DIAGNOSIS OF VIRAL HEPATITIS

Investigations

- ALT, AST, Bilirubin (Direct/Indirect), Glucose, INR
- Rule out: drug-induced, acute cholestatic injury, vascular pathology

Serologic/viral markers

HAV:
- IgG: acute or resolved infection
- IgM: acute infection

HBV:
- HBsAg: acute or chronic infection (if positive)
- HBeAg: marker of disease activity
- Anti-HBs: immunity
- Anti-HBc IgM: acute infection or exacerbation
- HBV DNA: viral activity

HCV:
- HCV RNA: viral activity
- Anti-HCV: antibodies to HCV

Note, autoimmune hepatitis and Wilson's disease may initially mimic acute viral hepatitis

RISK FACTORS

- Location of birth (HepB)
- Incarceration (HepB)

- Injection drug use (HepB, C, D)
- High-risk sexual behavior (HepB, C, D)

- Homelessness (HepA, B)
- Daycare exposure (HepA, E)

- Travel history (HepA, B, E>>> C, D)

EXAMPLES OF DRUG-RELATED CAUSES OF HEPATITIS

- Acetaminophen and NSAIDs
- Herbal products (e.g. buckthorn, chaparral, germander, black cohosh, nutmeg, valerian)
- Statins
- Antibiotic (e.g. amoxicillin-clavulanate, macrolides, minocycline, sulfamethoxazole-trimethoprim)
- Tumor Necrosis Factor (TNF) alpha inhibitors (e.g. infliximab)
- Interferon alpha/beta

GOALS OF THERAPY

- ✓ Prevent hepatitis infection
- ✓ Prevent liver cirrhosis, hepatocellular carcinoma (HBV/HCV) and hepatic failure (all)
- ✓ Cure the infection (HCV)
- ✓ Prevent further transmission of infection
- ✓ Identify drug-induced causes

PREVENTION OF VIRAL HEPATITIS

Vaccination is the optimal prevention strategy for HAV/HBV

- HBV vaccination confers immunity against HDV co-infection

Avoidance of risk-taking behaviours and harm reduction can assist in prevention of all types of hepatitis

ACUTE VIRAL HEPATITIS

- Acute viral hepatitis: viral infection that has been present for <6 months and results in signs of hepatocellular damage
- Symptoms:
  - Jaundice
  - Fever
  - Arthralgias/Myalgias
  - Headache
  - Fatigue/Malaise
  - Upper Right Quadrant Pain
  - Dark Urine
  - Pale Stools
- May cause fulminant hepatic failure (higher risk in pregnancy and patients with pre-existing liver dysfunction, co-infection)

ACUTE HEPATITIS: NON-PHARMACOLOGIC THERAPY

Alcohol consumption should be avoided for at least 3 months or until liver enzymes and hepatic function are completely normalized

No dietary or physical activity restrictions are necessary

Stop other hepatotoxic medications/herbals

ACUTE HEPATITIS: PHARMACOLOGIC THERAPY

Stop all hepatotoxic medications

Patients with INR > 1.4: give vitamin K 10 mg PO/SC

Patients with encephalopathy/possible fulminant liver failure: give lactulose targeting 2-3 loose bowel movements daily, consider rifaximin and referral to transplant center

In acute HBV infections, consider antiviral therapy with nucleos(t)ide analogues to prevent progression to liver failure and death

Most cases of acute hepatitis A, B, and E are self-limiting and patients recover without chronic complications

75% of patients with acute HCV will develop a chronic infection

PATHOPHYSIOLOGY: CHRONIC HEPATITIS

- Chronic viral hepatitis: virus present ≥6 months after infection
- Most commonly caused by HBV (presence of HBsAg) and HCV
  - HAV does not cause chronic infection
- Can progress to cirrhosis, hepatocellular carcinoma (HCC), and liver failure
- Co-infection is associated with more severe disease

NON-PHARMACOLOGIC THERAPY: CHRONIC HBV

Avoid excessive alcohol consumption (>20 g/day in women and >30 g/day in men)

No Encourage weight loss in patients with BMI >30 kg/m²

Optimize blood sugar control in patients with diabetes

Encourage smoking cessation

PHARMACOLOGIC THERAPY: CHRONIC HBV

25% of Chronic HBV will develop HCC and/or cirrhosis…but which 25%?

- Older patients (>40)
- Males
- Immunocompromised patients
- Patients with higher HBV DNA viral loads
- Elevated ALT
- Prolonged HBeAg seroconversion
- Viral coinfections
- Other liver disease
- Ethanol use
- African/Asian ancestry

PHARMACOLOGIC THERAPY: CHRONIC HBV

Who gets treatment?

- Those likely to progress to cirrhosis*
- Those with HIV co-infection
- Those with HBV related complications (cirrhosis, renal failure) especially with HBV DNA > 2x10^3 IU/mL

DRUGS USED IN CHRONIC HEPATITIS B

Doses are FYI only

Drug	Indication, Dose	Side Effects	Comments
Peginterferon alfa-2a (Pegasys®)	Antiviral and Immunomodulator used to treat patients with low HBV DNA and high ALT (generally HBeAg+ phase 2). 180 µg once weekly SC × 48 wk	Flu-like syndrome, bacterial infections, fatigue, fever, muscle aches, asthenia, weight loss, diarrhea, nausea, headaches, irritability, depression, anxiety, difficulty concentrating and sleeping, memory loss, hair loss, retinopathy (particularly in those with pre-existing hypertension), soreness and redness at injection site, neutropenia and thrombocytopenia	• Neutropenia and thrombocytopenia managed by dose reductions • Neutropenia associated with peginterferon does not appear to increase susceptibility to infection • Contraindications: • Pregnancy, decompensated liver disease, HIV, neutropenia, severe cardiac disease, solid organ transplant (except liver), renal failure ongoing/untreated, alcohol abuse or IVDU, severe/untreated major depression or psychosis

DRUGS USED IN CHRONIC HEPATITIS B

Drug	Indication, Dose	Side Effects	Comments
Nucleoside Analogues	Entecavir (Baraclude®) First-line • Nucleoside-naïve patients: 0.5 mg once daily PO • Lamivudine-experienced patients: 1 mg once daily PO • Take on empty stomach	Increased aminotransferase levels (ALT), lactic acidosis	• Do not use nucleoside analogues as monotherapy for HBV in patients with HIV coinfection • Acute HBV disease exacerbations (marked ALT elevations) have been reported after nucleoside analogue discontinuation • High drug resistance associated with long-term use of lamivudine, monitor ALT and HBV DNA Q3-6 months • Avoid entecavir if lamivudine resistance • Entecavir reduces HBV DNA levels more than lamivudine • Use lamivudine for disease reactivation prophylaxis in patients on immunosuppressants (HBs-Ag- & anti-HBC+) • Associated with reductions in hepatic inflammation, increased HBe seroconversion rates and reduced risk for development of hepatocellular carcinoma
	Lamivudine • 100 mg once daily PO	Uncommon	

DRUGS USED IN CHRONIC HEPATITIS B

Drug	Indication, Dose	Side Effects	Comments
Nucleotide Analogues	Tenofovir disoproxil fumarate (Viread®) First-line • 300 mg once daily PO • Adjust frequency in renal impairment	Renal toxicity, Fanconi syndrome, lactic acidosis, syndrome, osteomalacia	• Used against HIV, HBV and lamivudine-resistant HBV • Do not use nucleoside analogues as monotherapy for HBV in patients with HIV coinfection • Acute HBV disease exacerbations (marked ALT elevations) have been reported after nucleoside analogue discontinuation • Avoid TDF or dose reduce in CrCl < 50 ml/min • Avoid TAF in CrCl < 15 ml/min • Tenofovir alafenamide has a high barrier to resistance • Associated with reductions in hepatic inflammation, increased HBe seroconversion rates and reduced risk for development of hepatocellular carcinoma
	Tenofovir alafenamide (Vemlidy®) First-line • 25 mg once daily PO • No renal adjustment	Lactic acidosis, severe hepatomegaly with steatosis	

PHARMACOLOGIC THERAPY: HBV IN PREGNANCY

- Avoid PEGIFN (contraindicated)
- If pregnant and high viral load (HBV DNA > 200.000 IU/mL), start TDF or TAF as soon as possible and no later than the 2nd trimester to reduce vertical transmission
- Close monitoring of LFT post partum due to risk of flares (ALT testing every 4 weeks for the first 3 months, then at 6 months, then routine monitoring afterwards.
- Children born to mothers with HBsAg+ should receive HBIg and HBV vaccine as soon as possible, within 12 hours of delivery (2^{nd} and 3^{rd} doses of HBV vaccine should be completed before 6 months)
- Breastfeeding is permissible in treated or untreated patients

MONITORING PARAMETERS: HEPATITIS B

Parameter	Degree of Change	Timeframe
Acute signs of hepatitis (elevated ALT)	Improve/resolve	Every 4 weeks for the first 3 months, then once every 3 months for 1 year, then once every 6 months afterwards, as per the routine standard of care
Chronic hepatitis	Achieve SVR (no HBV DNA) Prevent cirrhosis and HCC Achieve HBeAg/HBsAg seroconversion	

CHRONIC HEPATITIS C

- In adults, 75–85% of acute HCV infections develop into chronic infections
- Risk factors for HCV:
    - Current/past IVDU (even once)
    - Healthcare access in areas without proper infection control
    - Blood transfusion or organ transplant before 1992 (in Canada)
    - History of incarceration
    - Place of birth with high endemic prevalence
        - Asia, Oceania, Eastern Europe, Africa/Middle East
    - Born to HCV infected mother
    - Sexual contact with someone who is HCV +
    - HIV infection
    - Hemodialysis
    - Elevated ALT NYD
    - Born between 1945-1975

CHRONIC HEPATITIS C

- In adults, 75–85% of acute HCV infections develop into chronic infections
- Risk factors for cirrhosis/fibrosis
    - Males
    - Age >40y at initial infection
    - Duration of infection
    - Alcohol consumption >50g/day
    - HIV co-infection
    - Immunosuppression
    - Diabetes
    - High BMI

CHRONIC HEPATITIS C

Investigations:

- Initial anti-HCV (antibody test) marks HCV exposure (not immunity)
- HCV RNA detection by PCR to confirm diagnosis
- Prior to treatment, assess for HCV RNA viral load, genotype, hepatic fibrosis
- HCV RNA test must be completed 12-24 weeks after treatment to determine status of sustained virologic response (SVR)

Patients with HCV should be tested for HIV coinfection

NON-PHARMACOLOGIC THERAPY

Limit alcohol consumption; moderate (1-19 g/day) and excessive (≥30 g/day) intake may increase risk of mortality

Optimize blood sugar control in patients with diabetes

Obese patients should consider weight loss strategies in order to reduce progression to liver fibrosis and cirrhosis

PHARMACOLOGIC THERAPY – CHRONIC HEPATITIS C

- Primary objective of chronic HCV therapy is complete eradication of the virus, termed sustained virological response (SVR)
- SVR is defined as:
  - Undetectable serum HCV RNA (<10-15 units/mL) 12 weeks post treatment
- Mainstay of therapy are oral direct-acting antivirals (DAA)
  - Combination of all oral DAAs have greatly improved HCV treatment
  - 90% cure rates and therapies for all genotypes, disease levels, prior treatment status
  - Once daily administration

PHARMACOLOGIC THERAPY: POLYMERASE INHIBITORS

- Sofosbuvir is the only nucleoside inhibitor of HCV polymerase (NS5B) available in Canada
- Must be administered as combination therapy
- Sofosbuvir is usually given in combination with ledipasvir, velpatasvir, and voxilaprevir
- Sofosbuvir is P-glycoprotein (Pgp) substrate
  - Caution for reduced therapeutic effect with Pgp inducers (e.g. rifampin)
  - Pgp inhibitors can be used with sofosbuvir or ledipasvir
- Dual or triple therapy with sofosbuvir is associated with higher cure rates compared to peginterferon + ribavirin dual therapy

PHARMACOLOGIC THERAPY

Protease (NS3) Inhibitors

- glecaprevir, and voxilaprevir
- Inhibits HCV replication and used in combination with polymerase and NS5A inhibitors
- CYP3A4 substrates and inhibitors, caution re: drug interactions

NS5A Inhibitors

- ledipasvir, pibrentasvir and velpatasvir
- Pgp substrates and CYP3A4 substrates (except ledipasvir)
- Minimal drug interactions, caution with statins, antituberculosis medications, HIV regimens
- Ledipasvir and velpatasvir require acid for absorption (separate from PPIs/antacids)

Ribavirin

- Antiviral medication that is only used in combination with other agents

HEPATITIS C TREATMENT

Resistance

- Antiviral resistance is a major concern
- Resistance occurs due to low fidelity of RNA-dependent RNA polymerase and high replication rate of HCV

Overcoming resistance

- Strategies to overcome resistance include:
  - Avoiding DAA monotherapy
  - DAA dose reductions
  - Maximizing adherence
  - Combining DAAs with non-overlapping resistance profiles
  - Choosing DAAs with high barriers to resistance
  - Combining DAAs with PEG-IFN and RBV

DRUGS USED IN CHRONIC HEPATITIS C: COMBINATION ANTIRETROVIRALS

Drug	Indication, Dose	Side Effects	Comments
Sofosbuvir/ Velpatasvir (Epclusa®)	• Sofosbuvir 400 mg/velpatasvir 100 mg once daily PO • Treatment naïve with or without cirrhosis: × 12 wk • Treatment naïve or experienced with decompensated cirrhosis (Child-Pugh score B or C): × 12 wk plus ribavirin	• Headache, fatigue • Requires acid for absorption, decreased action if used with acid-reducing agents • Bradycardia if used with amiodarone	• Used for genotype 1, 2, 3, 4, 5, 6 (note: Harvoni does not cover genotype 2 or 3) • Take with or without food • Risk of hep B reactivation if HBV and HCV co-infection ◦ Check HBV status prior to treatment • Important to review patient's medication history for drug interactions
Ledipasvir/ Sofosbuvir (Harvoni®)	• Ledipasvir 90 mg/sofosbuvir 400 mg once daily PO • Treatment duration depends on genotype, cirrhosis, and treatment experience		

DRUGS USED IN CHRONIC HEPATITIS C: COMBINATION ANTIRETROVIRALS

Drug	Indication, Dose	Side Effects	Comments
Glecaprevir/ Pibrentasvir (Maviret®)	• Glecaprevir 300 mg/pibrentasvir 120 mg once daily PO • Treatment duration depends on genotype, cirrhosis, and treatment experience	Headache, fatigue	• Used for genotype 1, 2, 3, 4, 5, 6 • May be used in patients without cirrhosis or with compensated cirrhosis, renal impairment (including dialysis-dependent patients), and HIV co-infection

DRUGS USED IN CHRONIC HEPATITIS C: COMBINATION ANTIRETROVIRALS

Drug	Indication, Dose	Side Effects	Comments
Sofosbuvir/ Velpatasvir/ Voxilaprevir (Vosevi®)	• Sofosbuvir 400 mg/velpatasvir 100 mg/voxilaprevir 100 mg once daily PO • Genotype 1, 2, 3, 4, 5, 6 without cirrhosis or with compensated cirrhosis: treatment experienced with NS5A inhibitor: ×12 wk • Genotype 1, 2, 3, 4 without cirrhosis or with compensated cirrhosis: treatment experienced with sofosbuvir (NS5A inhibitor naïve): ×12 wk	• Headache, fatigue, diarrhea, nausea, insomnia, decreased energy/strength • Requires acid for absorption, decreased action if used with acid-reducing agents • Bradycardia if used with amiodarone	• Used for genotype 1, 2, 3, 4, 5, 6 • Take with food • Approved for prior DAA treatment failures • Risk of hep B reactivation if HBV and HCV co-infection • Check HBV status prior to treatment • Important to review patient's medication history for drug interactions

DRUGS USED IN CHRONIC HEPATITIS C

Drug	Indication, Dose	Side Effects	Comments
Ribavirin	<75 kg: 1000 mg daily PO in 2 divided doses ≥75 kg: 1200 mg daily PO in 2 divided doses	Dry cough, rash, hemolytic anemia (resulting in worsening of cardiac disease and may lead to fatal/nonfatal MI)	• Ribavirin is always used in combination with other agents and is ineffective as monotherapy • Contraindicated in pregnant patients and their male partners • Avoid use in patients with CrCl<50mL/min • Manage hemolytic anemia with dose reductions

MONITORING PARAMETERS – HEPATITIS C

Parameter	Degree of Change	Timeframe
Acute signs of hepatitis (elevated AST/ALT)	Improve/resolve	2-6 weeks (max. 6 months)
Chronic hepatitis	Eradicate infection and achieve SVR Prevent cirrhosis and liver complications	12-24 weeks

THERAPEUTIC TIPS

- Presence of HIV concurrently with HCV increases the probability of sexual transmission of HCV (low sexual transmission rate in heterosexual monogamous couples)
- Refer HIV and HCV/HBV co-infection cases to infectious disease specialists
- Consult transplant hepatologists in HBV/HCV patients with decompensated disease or history of liver transplant
- Key to treatment success is medication adherence and prompt follow-up
- Significant drug-drug interactions, especially with HIV coinfection

HEPATITIS A/B IMMUNIZING AGENTS – APPENDIX 1

FYI

Class	Vaccine	Indication
Passive Immunizing Agents	Hepatitis B immune globulin (HyperHEP B® S/D)	Postexposure prophylaxis for hepatitis B: 0.06 mL/kg IM Combined with active hep B vaccine for short & long-term protection Indicated for: • acute exposure to blood containing HBsAg • patients unwilling to vaccinate with Hep B or those who were not protected against HBV prior to HBV exposure • perinatal exposure of infants born to HBsAg+ mothers • sexual exposure to an HBsAg+ person • household exposure to persons with acute HBV infection
	Pooled immunoglobulins with activity against hep A (GamaSTAN® S/D)	Consider postexposure prophylaxis for hep A within 2w of exposure (if exposure is continuous, repeat in 5 months) Consider prophylaxis when: • Hepatitis A vaccine is unavailable, CI, or not used prior to exposure • Age <1 yr, immunocompromised, or chronic liver disease

HEPATITIS A/B IMMUNIZING AGENTS – APPENDIX 1

FYI

Class	Vaccine	Dose
Viral Vaccines (combinations)	Combined hepatitis A and B (Twinrix®, Twinrix® Junior)	Twinrix® for adults ≥19 y: IM at 0, 1 and 6 months or × at 0, 7, 21 and 365 days Twinrix® Junior for children 6 months to <19 y: 0.5 mL IM at 0, 1 and 6 months
Viral Vaccines (single component)	Hepatitis A vaccine (Avaxim®, Havrix® 1440, Havrix® 720 Junior, Vaqta®)	Avaxim® for adults and children ≥12 y: 0.5 mL (160 AU) IM Havrix 1440 for adults ≥19 y: 1 mL IM Havrix 720 Junior for children 1–18 y: 0.5 mL IM All above given at 0 and 6–12 months Vaqta® for adults ≥18 y: 1 mL IM and booster dose 6-18 months later; pediatric dosing for children 12 months–17 y: 0.5 mL IM given at 0 and 6-18 months

HEPATITIS A/B IMMUNIZING AGENTS – APPENDIX 1

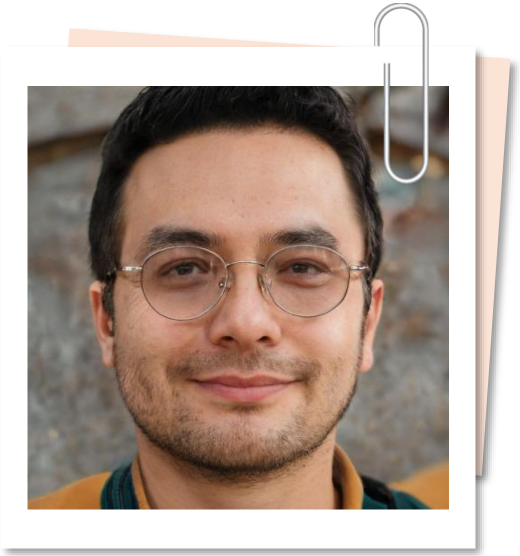
Class	Vaccine	Dose
Viral Vaccines (single component)	Hepatitis B vaccine (Engerix-B, Recombivax HB®)	Engerix-B: Adults ≥20 y: 1 mL (20 µg HBsAg) IM at 0, 1 and 6 months OR at 0, 1, 2 and 12 months OR at 0, 7, 21 and 365 days Adults (dialysis ≥16 y): 2 mL IM at 0, 1, 2 and 6 months Children 0–19 y: 0.5 mL IM at 0, 1 and 6 months OR at 0, 1, 2 and 12 months Children 11–15 y (alternative 2-dose schedule): 1 mL IM at 0 and 6 months Immunocompromised: 2 mL IM. Monitor anti-HBs yearly and give boosters PRN Recombivax HB®: Adults ≥20 y: 1 mL (10 µg HBsAg) IM at 0, 1 and ≥2 months Children 11–19 y: 0.5 mL IM at 0, 1 and ≥2 months Children 11–15 y (alternative 2-dose schedule): 1 mL IM at 0 and 4–6 months Infants and children 0 to <11 y: 0.25 mL IM at 0, 1 and ≥2 months

CASE 1

FY is a 41-year-old male presenting to your hepatology clinic for an initial appointment. FY was referred to your clinic by his rheumatologist after having routine blood work before starting a TNF α inhibitor for psoriatic arthritis. FY has a past medical history of psoriatic arthritis (diagnosed at age 37) and CKD (baseline CrCl 44 mL/min). He currently uses betamethasone valerate 0.05% cream applied BID and takes sulfasalazine 500 mg PO BID, and naproxen 500 mg PO TID. He has no allergies to medications. He currently works in IT. He is in a monogamous relationship (MSM). He drinks alcohol socially and uses cannabis occasionally.

His results reveal the following:

- HBsAg +
- HBsAb –
- HBcAb +
- ALT 49 U/L (normal value 7-40 U/L)
- HIV negative



CASE 1 - QUESTIONS

1. Based on the serology results, what is the likely diagnosis for FY?
   a) Acute hepatitis B
   b) Resolved hepatitis B
   c) Chronic hepatitis B
   d) Vaccine induced hepatitis B immunity

2. Based on the question above, which of the following best reflect the risk FY is encountering when starting a TNFα inhibitor?
   a) Reactivation of hepatitis B leading to ascites
   b) Increased hepatitis B viral replication
   c) Worsening control of psoriatic arthritis
   d) Hepatocellular carcinoma (HCC) from hepatitis B reactivation

3. Which of the following would be vital in determining the optimal approach to pharmacotherapy for FY?
   a) HBV viral load
   b) AST
   c) HBeAg
   d) SCr

4. What would you recommend FY take to treat his condition?
   a) Tenofovir disoproxil fumarate
   b) Lamivudine
   c) PEG-interferon
   d) Tenofovir alafenamide

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CHANGE LOG

July 2025

- Formatting changes made; no change to content

October 2025

- Slide 35: Removed brand name Ibavyr® since it's dormant on the Canadian market

March 2026

- Slide 2: Updated NAPRA competencies
- Minor formatting changes
- Removed slide 20 as adefovir is not available on the Canadian market anymore
- Slide 21: Updated information
- Slide 22: Updated monitoring parameters
- Slide 31: Added that Harvoni® does not cover genotype 3 (in addition to 2)
- Added FYI to slides 37 and 38
- Updated references

April 2026

- Slide 32: Updated information on Maviret® to indicate it can be used in patients without cirrhosis or with compensated cirrhosis